Chapter 7. The Zeta Function and Prime Number Theorem

Zeros of ζ on $\operatorname{Re} s = 1$, Chebyshev functions, proof of the prime number theorem

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Overview

Euler's product formula for ζ connects analysis with the arithmetic of primes; an easy consequence is that $\sum_p 1/p$ diverges. To understand the distribution of primes, let

$$\pi(x) = \text{number of primes less than or equal to } x.$$

Legendre and Gauss conjectured

$$\pi(x) \sim \frac{x}{\log x} \qquad \text{as } x \to \infty, \tag{1}$$

where $f(x) \sim g(x)$ means $f(x)/g(x) \to 1$. Tchebychev proved by elementary methods the weaker $\pi(x) \approx x/\log x$, i.e. $A\frac{x}{\log x} \le \pi(x) \le B\frac{x}{\log x}$ for large x (2). In 1896 Hadamard and de la Vallée Poussin proved (1), the *prime number theorem*, using complex analysis, as we do. At the heart of the proof lies the fact that $\zeta(s)$ does not vanish on the line $\mathrm{Re}(s) = 1$ (the two statements are in fact equivalent).

1 Zeros of the zeta function

Euler's identity. For $\mathrm{Re}(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$

Proof. Each factor is a convergent geometric series $1 + p^{-s} + p^{-2s} + \cdots$; formally multiplying them gives the result. Precisely, take first $s > 1$ real and integers $M > N$. By unique factorization, every $n \le N$ is a product of primes $\le N$, each repeated fewer than M times, so

$$\sum_{n=1}^{N} \frac{1}{n^s} \le \prod_{p \le N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots + \frac{1}{p^{Ms}}\right) \le \prod_{p \le N} \frac{1}{1 - p^{-s}} \le \prod_p \frac{1}{1 - p^{-s}},$$

and letting $N \to \infty$, $\sum_{n=1}^{\infty} n^{-s} \le \prod_p (1 - p^{-s})^{-1}$. Conversely, again by unique factorization, $\prod_{p \le N}(1 + p^{-s} + \cdots + p^{-Ms}) \le \sum_{n=1}^{\infty} n^{-s}$; letting $M \to \infty$, then $N \to \infty$, gives the reverse inequality. (For complex s with $\mathrm{Re}(s) > 1$ the identity follows by analytic continuation, or by the same argument with absolute convergence.) ∎

By Proposition 3.1, $\zeta(s) \neq 0$ for $\mathrm{Re}(s) > 1$. The functional equation $\xi(s) = \xi(1 - s)$ reads $\pi^{-s/2}\Gamma(s/2)\zeta(s) = \pi^{-(1-s)/2}\Gamma((1-s)/2)\zeta(1-s)$, so

$$\zeta(s) = \pi^{s-1/2}\frac{\Gamma((1-s)/2)}{\Gamma(s/2)}\zeta(1-s).$$

For $\mathrm{Re}(s) < 0$: (i) $\zeta(1-s)$ has no zeros, as $\mathrm{Re}(1-s) > 1$; (ii) $\Gamma((1-s)/2)$ is zero free; (iii) $1/\Gamma(s/2)$ has zeros at $s = -2, -4, -6, \ldots$. Hence:

Theorem 1.1

The only zeros of ζ outside the strip $0 \le \mathrm{Re}(s) \le 1$ are at the negative even integers, $-2, -4, -6, \ldots$.

The region $0 \le \mathrm{Re}(s) \le 1$ is the **critical strip**; the zeros outside it are the **trivial zeros**. Riemann conjectured the **Riemann hypothesis**: *the zeros of $\zeta(s)$ in the critical strip lie on the line* $\mathrm{Re}(s) = 1/2$—still one of the most famous unsolved problems. We prove:

Theorem 1.2

The zeta function has no zeros on the line $\mathrm{Re}(s) = 1$.

(Since ζ has a pole at 1 there are no zeros near $s = 1$; the point is the deeper fact $\zeta(1 + it) \neq 0$ for all $t \in \mathbb{R}$.) By the functional equation, ζ then has no zeros on $\mathrm{Re}(s) = 0$ either.

Lemma 1.3

If $\mathrm{Re}(s) > 1$, then

$$\log \zeta(s) = \sum_{p,m} \frac{p^{-ms}}{m} = \sum_{n=1}^{\infty} c_n n^{-s}$$

for some $c_n \geq 0$.

Proof. For $s > 1$, take the logarithm of Euler's product and use $\log \frac{1}{1-x} = \sum_{m=1}^{\infty} \frac{x^m}{m}$ $(0 \leq x < 1)$:

$$\log \zeta(s) = \sum_{p} \log\Big(\frac{1}{1 - p^{-s}}\Big) = \sum_{p,m} \frac{p^{-ms}}{m}.$$

The double sum converges absolutely, so the order of summation is irrelevant (see the Note at the end of the chapter). The formula holds for $\mathrm{Re}(s) > 1$ by analytic continuation; $\log \zeta$ is well defined in the simply connected half-plane $\mathrm{Re}(s) > 1$ since ζ has no zeros there (Theorem 6.2). Finally $c_n = 1/m$ if $n = p^m$ and $c_n = 0$ otherwise. ∎

Lemma 1.4

If $\theta \in \mathbb{R}$, then $3 + 4\cos\theta + \cos 2\theta \geq 0$.

This follows from $3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2$.

Corollary 1.5

If $\sigma > 1$ and t is real, then $\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| \geq 0$.

Proof. With $s = \sigma + it$, $\mathrm{Re}(n^{-s}) = \mathrm{Re}(e^{-(\sigma+it)\log n}) = n^{-\sigma}\cos(t \log n)$. Hence, with $\theta_n = t \log n$,

$$\begin{aligned}\log|\zeta^3(\sigma)\zeta^4(\sigma + it)\zeta(\sigma + 2it)| &= 3\,\mathrm{Re}[\log \zeta(\sigma)] + 4\,\mathrm{Re}[\log \zeta(\sigma + it)] + \mathrm{Re}[\log \zeta(\sigma + 2it)] \\ &= \sum c_n n^{-\sigma}(3 + 4\cos\theta_n + \cos 2\theta_n) \geq 0,\end{aligned}$$

by Lemma 1.4 and $c_n \geq 0$. ∎

Proof of Theorem 1.2. Suppose $\zeta(1 + it_0) = 0$ for some $t_0 \neq 0$. ζ is holomorphic at $1 + it_0$ and vanishes there to order ≥ 1, so $|\zeta(\sigma + it_0)|^4 \leq C(\sigma - 1)^4$ as $\sigma \to 1$. Since $s = 1$ is a simple pole, $|\zeta(\sigma)|^3 \leq C'(\sigma - 1)^{-3}$ as $\sigma \to 1$. And $|\zeta(\sigma + 2it_0)|$ stays bounded as $\sigma \to 1$. Hence $|\zeta^3(\sigma)\zeta^4(\sigma + it_0)\zeta(\sigma + 2it_0)| \to 0$ as $\sigma \to 1^+$, contradicting Corollary 1.5 (the logarithm of a number in $(0, 1)$ is negative). ∎

1.1 Estimates for $1/\zeta(s)$

The proof uses the logarithmic derivative ζ'/ζ near $\mathrm{Re}(s) = 1$, so we need growth estimates for ζ' (Proposition 2.7) and for $1/\zeta$. The following is a quantitative version of Theorem 1.2.

Proposition 1.6

For every $\epsilon > 0$, we have $1/|\zeta(s)| \leq c_\epsilon |t|^\epsilon$ when $s = \sigma + it$, $\sigma \geq 1$, and $|t| \geq 1$.

Proof. From Corollary 1.5, $|\zeta^3(\sigma)\zeta^4(\sigma+it)\zeta(\sigma+2it)| \geq 1$ for $\sigma \geq 1$. With the estimate for ζ in Proposition 2.7 and $|\zeta(\sigma)| \leq c/(\sigma-1)$,

$$|\zeta^4(\sigma+it)| \geq c|\zeta^{-3}(\sigma)||t|^{-\epsilon} \geq c'(\sigma-1)^3|t|^{-\epsilon},$$

so

$$|\zeta(\sigma+it)| \geq c'(\sigma-1)^{3/4}|t|^{-\epsilon/4} \qquad (\sigma \geq 1,\ |t| \geq 1). \tag{3}$$

Two cases, with a constant A to be chosen. If $\sigma - 1 \geq A|t|^{-5\epsilon}$, (3) gives $|\zeta(\sigma+it)| \geq A'|t|^{-4\epsilon}$, and replacing 4ϵ by ϵ gives the estimate. If $\sigma - 1 < A|t|^{-5\epsilon}$, choose $\sigma' > \sigma$ with $\sigma' - 1 = A|t|^{-5\epsilon}$. Then $|\zeta(\sigma+it)| \geq |\zeta(\sigma'+it)| - |\zeta(\sigma'+it) - \zeta(\sigma+it)|$, and by the mean value theorem and the bound on ζ' (Proposition 2.7), $|\zeta(\sigma'+it) - \zeta(\sigma+it)| \leq c''|\sigma' - \sigma||t|^{\epsilon} \leq c''(\sigma'-1)|t|^{\epsilon}$. With (3) at σ',

$$|\zeta(\sigma+it)| \geq c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} - c''(\sigma'-1)|t|^{\epsilon}.$$

Choose $A = (c'/(2c''))^4$; since $\sigma' - 1 = A|t|^{-5\epsilon}$, this makes $c'(\sigma'-1)^{3/4}|t|^{-\epsilon/4} = 2c''(\sigma'-1)|t|^{\epsilon}$, hence $|\zeta(\sigma+it)| \geq c''(\sigma'-1)|t|^{\epsilon} = A''|t|^{-4\epsilon}$. Replace 4ϵ by ϵ. ∎

2 Reduction to the functions ψ and ψ_1

Tchebychev introduced a function easier to handle than $\pi(x)$ but carrying the same information:

$$\psi(x) = \sum_{p^m \leq x} \log p,$$

summed over integers of the form $p^m \leq x$ (p prime, $m \geq 1$). Equivalently, with $\Lambda(n) = \log p$ if $n = p^m$ for some prime p and $m \geq 1$, and $\Lambda(n) = 0$ otherwise,

$$\psi(x) = \sum_{1 \leq n \leq x} \Lambda(n), \qquad \psi(x) = \sum_{p \leq x} \left[\frac{\log x}{\log p}\right] \log p,$$

$[u]$ the greatest integer $\leq u$ (since $p^m \leq x$ iff $m \leq \log x/\log p$).

Proposition 2.1

If $\psi(x) \sim x$ as $x \to \infty$, then $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. We show $1 \leq \liminf \pi(x)\frac{\log x}{x}$ and $\limsup \pi(x)\frac{\log x}{x} \leq 1$ (4). Crudely, $\psi(x) = \sum_{p \leq x}\left[\frac{\log x}{\log p}\right]\log p \leq \sum_{p \leq x}\frac{\log x}{\log p}\log p = \pi(x)\log x$, so $\frac{\psi(x)}{x} \leq \frac{\pi(x)\log x}{x}$, and $\psi(x) \sim x$ gives the first inequality. For the second, fix $0 < \alpha < 1$:

$$\psi(x) \geq \sum_{p \leq x} \log p \geq \sum_{x^\alpha < p \leq x} \log p \geq (\pi(x) - \pi(x^\alpha))\alpha \log x,$$

so $\psi(x) + \alpha\pi(x^\alpha)\log x \geq \alpha\pi(x)\log x$. Dividing by x, and using $\pi(x^\alpha) \leq x^\alpha$, $\alpha < 1$, $\psi(x) \sim x$, we get $1 \geq \alpha \limsup \pi(x)\frac{\log x}{x}$. As $\alpha < 1$ is arbitrary, the proof is complete. ∎

(The converse also holds.) It is more convenient to work with

$$\psi_1(x) = \int_1^x \psi(u)\,du.$$

Proposition 2.2

If $\psi_1(x) \sim x^2/2$ as $x \to \infty$, then $\psi(x) \sim x$ as $x \to \infty$, and therefore $\pi(x) \sim x/\log x$ as $x \to \infty$.

Proof. By Proposition 2.1 it suffices to show $\psi(x) \sim x$. Since ψ is increasing, for $\alpha < 1 < \beta$,

$$\frac{1}{(1-\alpha)x}\int_{\alpha x}^{x} \psi(u)\,du \le \psi(x) \le \frac{1}{(\beta-1)x}\int_{x}^{\beta x} \psi(u)\,du.$$

So $\psi(x) \le \frac{1}{(\beta-1)x}[\psi_1(\beta x) - \psi_1(x)]$, i.e. $\frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{\psi_1(\beta x)}{(\beta x)^2}\beta^2 - \frac{\psi_1(x)}{x^2}\Big]$, whence

$$\limsup_{x\to\infty} \frac{\psi(x)}{x} \le \frac{1}{\beta-1}\Big[\frac{1}{2}\beta^2 - \frac{1}{2}\Big] = \frac{1}{2}(\beta+1).$$

As $\beta > 1$ is arbitrary, $\limsup \psi(x)/x \le 1$; a similar argument with $\alpha < 1$ gives $\liminf \psi(x)/x \ge 1$. ∎

To relate ψ_1 and ζ: differentiating Lemma 1.3, for $\mathrm{Re}(s) > 1$, $\frac{\zeta'(s)}{\zeta(s)} = -\sum_{m,p}(\log p)p^{-ms} = -\sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}$, i.e.

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty}\frac{\Lambda(n)}{n^s}. \tag{5}$$

Proposition 2.3

For all $c > 1$,

$$\psi_1(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)\,ds. \tag{6}$$

Lemma 2.4

If $c > 0$, then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{a^s}{s(s+1)}\,ds = \begin{cases} 0 & \text{if } 0 < a \le 1,\\ 1 - 1/a & \text{if } 1 \le a.\end{cases}$$

Here, the integral is over the vertical line $\mathrm{Re}(s) = c$.

Proof. $|a^s| = a^c$ on the line, so the integral converges. Suppose $a \ge 1$, $a = e^{\beta}$, $\beta = \log a \ge 0$, and $f(s) = \frac{a^s}{s(s+1)} = \frac{e^{\beta s}}{s(s+1)}$; then $\mathrm{res}_{s=0} f = 1$ and $\mathrm{res}_{s=-1} f = -1/a$. For $T > 0$ let $\Gamma(T)$ be the vertical segment $S(T)$ from $c - iT$ to $c + iT$ together with the half-circle $C(T)$ of radius T centered at c, to the left of the segment, positively oriented (a toy contour). For T large, 0 and -1 are inside, and $\frac{1}{2\pi i}\int_{\Gamma(T)} f = 1 - 1/a$. On $C(T)$, for large T, $|s(s+1)| \ge \frac{1}{2}T^2$ and (as $\sigma \le c$) $|e^{\beta s}| \le e^{\beta c}$, so $\big|\int_{C(T)} f\big| \le \frac{C}{T^2}2\pi T \to 0$. If $0 < a \le 1$, use the half-circle to the right of $\mathrm{Re}(s) = c$; there are no poles inside and the same estimate applies (now $\beta \le 0$ and $\sigma \ge c$). ∎

Proof of Proposition 2.3. $\psi(u) = \sum_{n=1}^{\infty}\Lambda(n)f_n(u)$ with $f_n(u) = 1$ if $n \le u$ and 0 otherwise. Hence

$$\psi_1(x) = \int_0^x \psi(u)\,du = \sum_{n=1}^{\infty}\Lambda(n)\int_0^x f_n(u)\,du = \sum_{n\le x}\Lambda(n)(x-n).$$

By (5) and Lemma 2.4 with $a = x/n$,

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big)ds = x\sum_{n=1}^{\infty}\Lambda(n)\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}\frac{(x/n)^s}{s(s+1)}\,ds = x\sum_{n\le x}\Lambda(n)\Big(1-\frac{n}{x}\Big) = \psi_1(x).$$

∎

2.1 Proof of the asymptotics for ψ_1

We show $\psi_1(x) \sim x^2/2$, which completes the proof of the prime number theorem. The ingredients: the formula (6); the non-vanishing $\zeta(1+it) \neq 0$; and the estimates of Proposition 2.7 and Proposition 1.6. Strategy: move the line of integration from $\mathrm{Re}(s) = c$ to $\mathrm{Re}(s) = 1$, so that x^{s+1} has size x^2 instead of x^{c+1}; the pole of ζ at $s = 1$ contributes exactly the main term $x^2/2$, and the rest must be shown to be essentially smaller.

Fix $c = 2$, $x \geq 2$, and let

$$F(s) = \frac{x^{s+1}}{s(s+1)}\Big(-\frac{\zeta'(s)}{\zeta(s)}\Big).$$

Deform the line $\mathrm{Re}(s) = c$ to the path $\gamma(T)$ ($T \geq 3$, to be chosen): along $\mathrm{Re}(s) = 1$ for $|t| \geq T$, and around $s = 1$ through the segment $\mathrm{Re}(s) = c$, $|t| \leq T$, joined by horizontal segments at height $\pm T$. Then

$$\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} F(s)\,ds = \frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds. \tag{7}$$

Indeed, by Proposition 2.7 and Proposition 1.6, $|\zeta'(s)/\zeta(s)| \leq A_\eta |t|^\eta$ for any fixed $\eta > 0$, when $\sigma \geq 1$, $|t| \geq 1$; so $|F(s)| \leq A'|t|^{-2+\eta}$ in the two infinite regions between the line and $\gamma(T)$, where F is regular, and Cauchy's theorem gives (7).

Next pass from $\gamma(T)$ to $\gamma(T,\delta)$, which consists of γ_1: $\mathrm{Re}(s) = 1$, $t \leq -T$; γ_2: horizontal segment at $t = -T$ from 1 to $1-\delta$; γ_3: $\mathrm{Re}(s) = 1-\delta$, $|t| \leq T$; γ_4: horizontal segment at $t = T$ from $1-\delta$ to 1; γ_5: $\mathrm{Re}(s) = 1$, $t \geq T$. For fixed T, choose $\delta > 0$ so small that ζ has no zeros in the box $\{\sigma + it : 1-\delta \leq \sigma \leq 1,\ |t| \leq T\}$ (possible since ζ does not vanish on $\sigma = 1$). F has a simple pole at $s = 1$: by Corollary 2.6, $\zeta(s) = \frac{1}{s-1} + H(s)$ with H regular near 1, so $-\zeta'(s)/\zeta(s) = \frac{1}{s-1} + h(s)$ with h holomorphic near 1, and $\mathrm{res}_{s=1} F = x^2/2$. Thus

$$\frac{1}{2\pi i}\int_{\gamma(T)} F(s)\,ds = \frac{x^2}{2} + \frac{1}{2\pi i}\int_{\gamma(T,\delta)} F(s)\,ds.$$

Let $\epsilon > 0$ be given. γ_1, γ_5. For $\sigma = 1$, $|x^{1+s}| = x^2$, and by Proposition 1.6 (with Proposition 2.7) $|\zeta'/\zeta| \leq A|t|^{1/2}$, so $\big|\int_{\gamma_5} F\big| \leq Cx^2 \int_T^\infty \frac{|t|^{1/2}}{t^2}\,dt$; the integral converges, so we can choose T with $\big|\int_{\gamma_1} F\big|, \big|\int_{\gamma_5} F\big| \leq \epsilon x^2$. γ_3. With T fixed, $|x^{1+s}| = x^{2-\delta}$ on γ_3, so $\big|\int_{\gamma_3} F\big| \leq C_T x^{2-\delta}$. γ_2, γ_4. $\big|\int_{\gamma_2} F\big| \leq C'_T \int_{1-\delta}^1 x^{1+\sigma}\,d\sigma \leq C'_T \frac{x^2}{\log x}$.

Hence there are constants C_T, C'_T with

$$\Big|\psi_1(x) - \frac{x^2}{2}\Big| \leq \epsilon x^2 + C_T x^{2-\delta} + C'_T \frac{x^2}{\log x}, \qquad \Big|\frac{2\psi_1(x)}{x^2} - 1\Big| \leq 2\epsilon + 2C_T x^{-\delta} + \frac{2C'_T}{\log x},$$

and for all large x, $\big|\frac{2\psi_1(x)}{x^2} - 1\big| \leq 4\epsilon$. So $\psi_1(x) \sim x^2/2$, and the prime number theorem is proved.

Note on interchanging double sums

Proposition

Let $\{a_{k\ell}\}_{1 \leq k,\ell < \infty}$ be complex numbers with

$$\sum_{k=1}^\infty \Big(\sum_{\ell=1}^\infty |a_{k\ell}|\Big) < \infty. \tag{8}$$

Then: (i) The double sum $A = \sum_{k=1}^\infty \sum_{\ell=1}^\infty a_{k\ell}$ converges, and the order of summation may be interchanged: $A = \sum_{\ell=1}^\infty \sum_{k=1}^\infty a_{k\ell}$. (ii) Given $\epsilon > 0$, there is N so that for all $K, L > N$,

$|A - \sum_{k=1}^{K}\sum_{\ell=1}^{L} a_{k\ell}| < \epsilon$. (iii) If $m \mapsto (k(m), \ell(m))$ is a bijection from $\mathbb{N}$ to $\mathbb{N} \times \mathbb{N}$ and $c_m = a_{k(m)\ell(m)}$, then $A = \sum_{m=1}^{\infty} c_m$.

Proof. (i) Let $b_k = \sum_\ell |a_{k\ell}|$, and let $S = \{x_0, x_1, \dots\}$ with $x_n \to x_0$. On S define $f_k(x_0) = \sum_{\ell=1}^{\infty} a_{k\ell}$, $f_k(x_n) = \sum_{\ell=1}^{n} a_{k\ell}$, and $g(x) = \sum_k f_k(x)$. Each f_k is continuous at x_0; $|f_k(x)| \le b_k$ with $\sum b_k < \infty$, so the series for g converges uniformly on S and g is continuous at x_0. Hence

$$\sum_k \sum_\ell a_{k\ell} = g(x_0) = \lim_n g(x_n) = \lim_n \sum_k \sum_{\ell=1}^{n} a_{k\ell} = \lim_n \sum_{\ell=1}^{n} \sum_k a_{k\ell} = \sum_\ell \sum_k a_{k\ell}.$$

(ii) $|A - \sum_{k\le K}\sum_{\ell\le L} a_{k\ell}| \le \sum_{k\le K}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>K}\sum_\ell |a_{k\ell}|$. Choose N with $\sum_{k>N} b_k < \epsilon/2$; the first sum is $\le \sum_{k\le N}\sum_{\ell>L}|a_{k\ell}| + \sum_{k>N} b_k$, and for L large the finitely many tails $\sum_{\ell>L}|a_{k\ell}|$, $k \le N$, are small (supplied). (iii) (Supplied.) Given ϵ, take N as in (ii) (for the absolute values as well), and M so large that $\{c_1, \dots, c_M\}$ contains all $a_{k\ell}$ with $k, \ell \le N$. For $M' \ge M$, $\sum_{m\le M'} c_m - \sum_{k,\ell\le N} a_{k\ell}$ is a sum of terms $a_{k\ell}$ with $k > N$ or $\ell > N$, of total absolute value $< \epsilon$; so $|\sum_{m\le M'} c_m - A| < 2\epsilon$. ∎